We can still return the `parents` dictionary to reconstruct the shortest paths to all the vertices that were reachable from `start`.

14.4.3 Analysis

While in BFS each vertex was added to the (plain) queue and never updated later, Dijkstra's algorithm uses a priority queue to keep track of the closest discovered vertices, and it’s possible that the priority[9] of a vertex changes after it was already added to the queue.

